

Problem

If $R = 20\ \Omega$, $L = 0.6\ \text{H}$, what value of C will make an RLC series circuit:

(a) overdamped?

(b) critically damped?

(c) underdamped?

Step-by-step solution

Step 1 of 6

Consider the series RLC circuit in Figure 1.

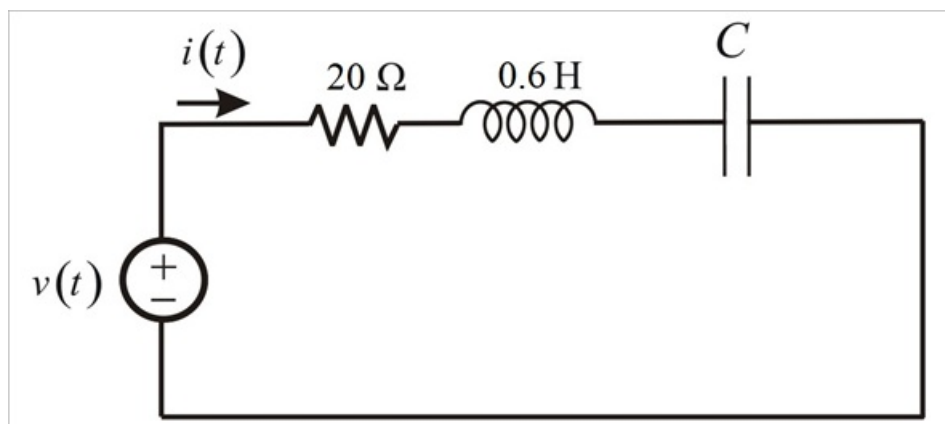


Figure 1

Step 2 of 6

Transform the given circuit from time domain to frequency domain.

Therefore L is transformed to sL , C is transformed $\frac{1}{sC}$, $i(t)$ is transformed to $I(s)$ and $v(t)$ is transformed to $V(s)$.

The frequency domain circuit is shown in Figure 2.

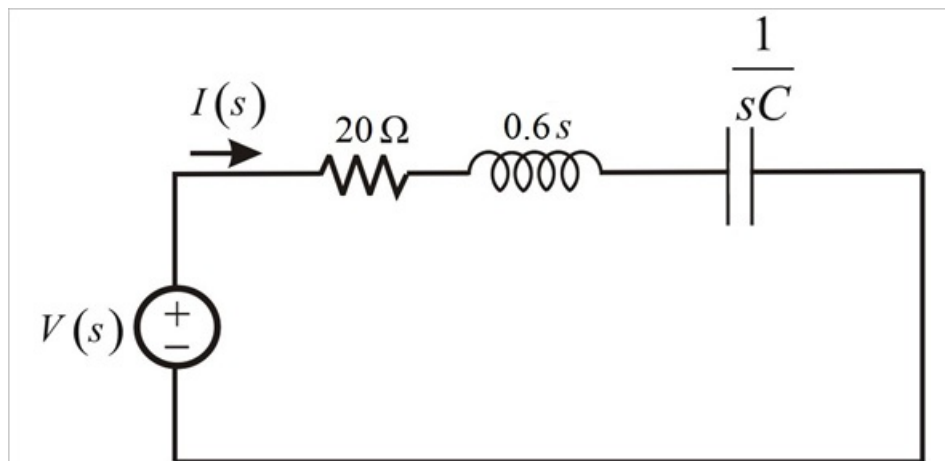


Figure 2

Step 3 of 6

From Figure 2, the current $I(s)$ in the circuit is,

$$\begin{aligned}
 I(s) &= \frac{V(s)}{20 + 0.6s + \frac{1}{sC}} \\
 &= \frac{V(s)}{20 + 0.6s + \frac{1}{sC}} \\
 &= \frac{sCV(s)}{0.6Cs^2 + 20Cs + 1} \\
 I(s) &= \frac{\frac{s}{0.6}V(s)}{s^2 + \frac{20}{0.6}s + \frac{1}{0.6C}} \dots\dots (1)
 \end{aligned}$$

Step 4 of 6

The characteristic equation of $I(s)$ is the denominator polynomial of equation (1) equals to zero.

$$s^2 + \frac{20}{0.6}s + \frac{1}{0.6C} = 0 \dots\dots (2)$$

Consider the characteristic equation as,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \dots\dots (3)$$

Here ζ is damping coefficient and ω_n is natural frequency.

Compare the equations (2) and (3).

$$2\zeta\omega_n = \frac{20}{0.6} \dots\dots (4)$$

$$\omega_n^2 = \frac{1}{0.6C} \dots\dots (5)$$

$$2\zeta\omega_n = \frac{20}{0.6}$$

$$2\zeta\omega_n = 33.33$$

$$4\zeta^2\omega_n^2 = 1111 \dots\dots (6)$$

Substitute equation (5) in equation (6).

$$4\zeta^2 \left(\frac{1}{0.6C} \right) = 1111$$

$$C = 6 \times 10^{-3} \zeta^2 \dots\dots (7)$$

Step 5 of 6

(a)

For over damped system, the damping coefficient $\zeta > 1$.

From equation (7).

Therefore the value of capacitance is $C > 6 \times 10^{-3} \text{ F}$.

(b)

For critically damped system, the damping coefficient $\zeta = 1$

From equation (7).

Therefore the value of capacitance is $C = 6 \times 10^{-3} \text{ F}$.

(c)

For under damped system, the damping coefficient $\zeta < 1$

From equation (7).

Therefore, the value of capacitance is $C < 6 \times 10^{-3} \text{ F}$.